Exponent Rules

Name	Rule	Example
Product Rule	$a^m \cdot a^n = a^{m+n}$	$4^2 \cdot 4^3 = 4^{(2+3)} = 4^5$
Quotient Rule	$\frac{a^m}{a^n} = a^{m-n}$	$\frac{5^3}{5^2} = 5^{(3-2)} = 5^1$
Power Rule	$(a^m)^n = a^{m \cdot n}$	$(2^4)^2 = 2^{4 \cdot 2} = 2^8$
Power of a Quotient	$\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}$	$\left(\frac{3}{5}\right)^3 = \frac{3^3}{5^3} = \frac{27}{125}$
Power of a Product	$(ab)^m = a^m \cdot b^m$	$(3 \cdot 5)^2 = 3^2 \cdot 5^2 = 9 \cdot 25 = 225$
Zero Rule	$a^0 = 1$	$113^0 = 1$
Negative Exponent	$a^{-m} = \frac{1}{a^m}$	$6^{-4} = \frac{1}{6^4}$
Fractional Exponents	$a^{\frac{m}{n}} = \sqrt[n]{a^m}$	$4^{\frac{2}{3}} = \sqrt[3]{4^2}$